

Crowding in School Choice[†]

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We consider the market design problem of matching students to schools in the presence of crowding effects. These effects are salient in parents' decision-making and the empirical literature; however, they cause difficulties in the design of satisfactory mechanisms and, as such, are not currently considered. We propose a new framework and an equilibrium notion that accommodates crowding, no-envy, and respect for priorities. The equilibrium has a student-optimal element that induces an incentive-compatible mechanism and is implementable via a novel algorithm. Moreover, analogs of fundamental structural results of the matching literature (the rural hospitals theorem, welfare lattice, etc.) survive. (JEL D47, H75, I21, I28)

Parents believe that if a school is overcrowded, then the quality of their children's education will suffer. This is supported by empirical literature. Card and Krueger (1992); Krueger and Whitmore (2001); Chetty et al. (2011); and Jackson, Johnson, and Persico (2016) all show that decreasing crowding, as measured by per capita expenditure or teacher-student ratios, etc., has positive effects on measures such as test scores, students' lifetime expected income, and career development. Policymakers also share parents' concerns. From 1999 to 2001, the United States' federal government allotted over \$4 billion to the Class-Size Reduction Program

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with the goal of smaller class sizes in grades K–3.¹ Similar programs exist in other countries.²

Matching theory has been astonishingly successful in practice, reforming the way students are assigned to schools in many municipalities worldwide. However, despite the salience of crowding effects, the theoretical school choice literature has largely avoided the issue. This is because whenever externalities are introduced into the standard model, its appealing features—like the existence of fair (in terms of respect for priorities) or stable matchings, the ordered structure of the set of such matchings, and the existence of incentive-compatible mechanisms—vanish (see Example 1 and the Related Literature section). As such, while students' preferences take into account crowding, the design of school choice mechanisms currently fails to do so.

The contribution of this paper is a novel framework to analyze the *school choice problem with crowding*. Our framework generalizes the now-standard model of Abdulkadiroğlu and Sönmez (2003) and restores its core features. There, each student has a preference over the finite set of schools. In our model, each student has a preference over the *two dimensions* of school identity *and* the total amount of educational resources that they consume at each school. The more crowded a school is, the fewer resources each student enjoys, and so the value of this second dimension at each school will emerge endogenously. The feature that allows us to cut the proverbial Gordian knot is modeling the level of educational resources as a continuous variable. This is, of course, an abstraction; however, the return on this abstraction is high—recovering existence of an ordered set of *fair* matchings, and a class of *fair* and *strategy-proof* rules. Our techniques also connect the school choice literature with research on Walrasian equilibria in the presence of price rigidities (see Section IIB). Finally, our framework naturally opens up new avenues for policy analysts. Eliciting preferences over crowding would provide powerful data for welfare analysis, and the model very naturally accommodates novel comparative statics like shifting resources between schools.

Case Study: Crowding in the Wake County Public School System.—Wake County is the most populous county in North Carolina and, in the last decade, was the second-fastest-growing in the United States among those with over a million residents.³ The Wake County Public School System (WCPSS) currently serves over 150,000 K–12 students. In 2016, the WCPSS projected the arrival of 32,000 new students, which indeed materialized into student population growth that has outpaced that of physical infrastructure. For example, in Abbotts Creek Elementary School for the 2019–2020 year, 870 students enrolled, or 141.2 percent of the school's total building capacity. This is not an uncommon phenomenon; Table 1⁴ lists crowding statistics for several other schools in the WCPSS, with some enrollments at almost twice the original building capacity. Student populations also fluctuate

¹ US Department of Education, Office of the Deputy Secretary, Policy and Program Studies Service, "A Descriptive Evaluation of the Federal Class-Size Reduction Program: Final Report," Washington DC, 2004.

² For example, the "Plus de Maîtres que de Classes" program in France in 2013 aimed to reduce class sizes in socioeconomically depressed areas (Réseaux d'Éducation Prioritaire).

³ US Census Bureau (2020).

⁴ Statistics from WCPSS (2020a) and (2020b).

TABLE 1. CROWDING STATISTICS FOR ELEMENTARY SCHOOLS IN WCPSS THAT ENACTED ENROLLMENT CAPS FOR THE 2020–2021 SCHOOL YEAR

	2018–2019	2019–2020				
	Population	Population	Physical capacity	Trailers	Crowding (%)	Crowding w/trailers (%)
Abbotts Creek	828	870	616	0	141.2	141.2
Alston Ridge	1,035	1,053	914	0	115.2	115.2
Beaverdam	778	790	616	3	128.2	116.9
Cedar Fork	999	848	508	8	166.9	126.9
Combs	858	767	410	12	187.1	118
Highcroft Drive	985	921	508	15	181.3	114
Holly Grove	977	940	814	3	115.5	107.6
Hortons Creek	988	911	716	0	127.2	127.2
Lead Mine	490	500	410	0	122	122
Mills Park	894	828	716	3	115.6	106.7
Oakview	909	901	716	5	125.8	110.4
Olive Chapel	984	978	612	13	159.8	112.2
Rogers Lane	861	730	616	4	118.5	104.9
Scotts Ridge	846	857	616	6	139.1	116.4
Sycamore Creek	1,148	1,082	914	0	118.4	118.4
Weatherstone	866	850	508	11	167.3	116.8
White Oak	596	747	616	0	121.3	121.3

Note: These amounted to 17 out of 120 elementary schools.

significantly from year to year. In Abbotts Creek, for the 2016–2017 year, there were 796 enrolled students, increasing to 870 for the 2019–2020 year and dropping to 854 in the following year.

The National Center for Education Statistics (NCES) also provides a wide range of data relevant to crowding, including enrollments, financial expenditure, number of teachers, and basic demographics on all public schools in the United States. The recently initiated NCES School-Level Finance Survey program collects and releases data on faculty and staff salaries, spending on supplies, spending on technology-related equipment, etc. For example, in 2016–2017 (latest available), Abbotts Creek had financial expenditures of \$3.5 million (or \$4,339 per student)(NCES 2017).

This case study immediately points to several items outside of the purview of the standard model. First, parents have access to data relevant to crowding and can condition their preferences over schools in this dimension. The figures above are publicly available via the WCPSS and NCES websites, as well as regularly reported on by the local media.⁵ Second, the fact that student enrollments frequently and grossly exceed building capacity challenges the standard assumption that enrollment caps are ex ante exogenously fixed. A school may almost always accommodate some extra students by adding seats in a classroom or a trailer. In the February 4, 2020, WCPSS Board of Education meeting, the assistant superintendent for school choice, planning, and assignment *proposed* enrollment caps

⁵ See, for example, Editorial Board, “More Growth Challenges Wake Schools,” *News and Observer*, March 26, 2016, accessed November 7, 2021, <https://www.newsobserver.com/opinion/editorials/article68451967.html>.

for each of the schools in Table 1 (among others)—after years of overcrowding.⁶ Many of the schools in Table 1 are therefore now considered *at* capacity. This then implies the third item, namely, that many schools are *below* capacity, with the potential for fluctuations year to year.⁷ The standard model would treat these all as simply unfilled schools, whose enrollment level does not affect welfare.

Our model addresses all of these points. We accommodate inviolable caps while providing a framework for analyzing the still-salient crowding levels at all schools, and in particular those below capacity. Parents are able to explicitly express preferences over such levels. Moreover, the *strategy-proof* and *fair* rules we propose make crowding levels a publicly announced quantity in the manner of an auctioneer, eliminating the need for parents to infer future crowding from the lagged data available.

The remainder of this section provides a sketch of the model and an overview of the formal results. Each student consumes a bundle consisting of a school and a level of educational resources. We require that all the students at a given school consume the same level, an important benchmark representing the ideal that all students are equally entitled to the resources of their school.⁸ An aggregate allocation is thus composed of a matching of students to schools and a vector of resource levels, one level for each school.

We propose a new equilibrium concept: rationing crowding equilibrium (RCE). The core of our innovation is in realizing that the vector of resource levels can function like a price. Consider a competitive solution applied in our context. We may imagine that an auctioneer announces a resource vector, which then determines a (finite) list of school and resource-level pairs. Each student will then demand (generically) one of these pairs, and we can ask the usual market clearing question: Does there exist a resource vector at which, for each school, the demand for educational resources is equal to its supply? We show that the answer is yes, if we allow for an error of at most *one* seat.^{9,10} Since each student faced the same budget set, the resulting allocation satisfies *no-envy*, at least for schools that have not reached their enrollment cap.

For schools constrained by their caps, our equilibrium incorporates *fairness* via respect for priorities. If student i prefers school s at an RCE, then school s has exhausted its capacity, and all students in that school have higher priority than student i . To continue the Walrasian analogy, a school exhausting capacity is tantamount to the auctioneer no longer being able to lower the announced resource

⁶See footnote 5. See also T. Keung Hui, “Wake County Puts Enrollment Caps On 19 Schools For 2020. One More May Be Added,” *News and Observer*, February 5, 2020, accessed November 7, 2021, <https://www.newsobserver.com/news/local/education/article239808038.html>.

⁷Apart from such proposals for enrollment caps, many states also have laws specifying a minimum teacher-student ratio and therefore implying such caps. For example, North Carolina requires at least one teacher per 18 students in grades K–3 (G.S. 115C-301 on “Allocation of teachers; class size”). Districts may also “manually” adjust capacities to accommodate unmatched students with specific requirements only available at certain schools, e.g., disabilities, large number of siblings.

⁸This is a restriction, and yet we have found an appealing solution to the problem. It follows that finding an appealing solution for the *unrestricted* case should be easier.

⁹More precisely, if a school at an RCE admits k students, each student is guaranteed to consume more than $1/(k + 1)$ fraction of the school’s total resources (see Remark 1).

¹⁰Excluding several specialized schools, the smallest (Wake STEM Early College) and largest (Apex Friendship) schools in Wake County in 2020–2021 had 256 and 2,733 students, respectively. In practice, then, the presence or not of a single seat at either school is negligible.

level of a school and thereby reduce demand, akin to a price ceiling.¹¹ If schools are below capacity, then assignments will be decided by our market-like mechanism without price rigidities.

Our main results are as follows. Fix a school choice problem with crowding. We show that an RCE exists for this problem so long as it satisfies a regularity condition that holds true generically (Theorem 1). We then establish a version of the Rural Hospitals theorem for our environment, i.e., in each RCE, each school is matched with the same number of students (Theorem 2). We show that the set of RCEs constitute a closed upper semilattice under the Pareto dominance quasi-order (Theorem 3), and so there exists a student-optimal RCE (Proposition 3). These are analogs of fundamental results in the literature, but the presence of crowding necessitates substantial difference in technique.

A *maximal RCE mechanism* recommends, for each problem in our domain, a maximal RCE (all in this set are welfare-equivalent and student-optimal). We show that these mechanisms are *strategy-proof* (Theorem 4), and we further find an algorithm to calculate them on a natural subdomain (Theorem 5).¹² It is in general hard to propose an algorithm for equilibrium computation when students' preferences are not quasi-linear (see the discussion after Theorem 1). This algorithm uses tools from both the multi-item auction (Demange, Gale, and Sotomayor 1986), with rationing constraints on students' demands, and the Deferred Acceptance algorithm (Gale and Shapley 1962). The latter adjusts the matching for schools whose capacity constraints have been met, and the former adjusts the distribution for unconstrained schools.

Discrete versus Continuous Crowding: Our Methodological Novelty

We model crowding via the consumption of a continuous good that comes in fixed supply. We have touted the benefits of this approach, weighed against the negligible “within-one-seat” relaxation of the market clearing condition. Thus, one might wonder if we could have achieved the same without such abstraction, by simply saying that if n students arrive at a school, then its aggregate resources are split n ways. We immediately face difficulties: two key solution concepts in the school choice literature, *no justified envy* and *nonwastefulness*, are incompatible. Recall that if student i envies student j , who is attending school s , then this envy is *justified* if school s prioritizes i over j . An allocation is *wasteful* if a student would want to switch to a school at which there is currently unfilled capacity.

EXAMPLE 1 (Incompatibility of No Justified Envy and Nonwastefulness): *Consider a simple problem with only two schools, s_1 and s_2 , and three students, 1, 2, and 3. Assume that s_1 and s_2 each have capacity three; each school also prioritizes student 3 over the others. Suppose student 1 has such strong preferences for s_1 that they prefer to attend s_1 regardless of the number of students there. Assume student 2 has similarly strong preferences for s_2 . Then at any nonwasteful allocation, we have*

¹¹ If the school admits further students, then crowding increases, and the resource level each student enjoys would decrease. So an upper bound on student enrollment is equivalent to a lower bound on the resource level.

¹² The general domain is infinite dimensional. We show the algorithm works on a finite-dimensional subdomain.

student 1 at s_1 and student 2 at s_2 . Finally, suppose student 3 would rather be alone than share either s_1 or s_2 with another student. If we put student 3 at s_1 , then since they have highest priority at s_2 , they have justified envy of student 2. A symmetric statement holds if we put student 3 at s_2 .

In work independent of ours, Copland (2021) studies exactly this model. They propose a natural relaxation of the standard solution concept: the envy that student 3 had for student 2 above is not yet justified unless student 3 is also willing to attend school s_2 , keeping all other students, including student 2, at their place.¹³ Given this definition, a solution always exists. They provide an algorithm to compute a solution, but it does not yield a *strategy-proof* mechanism. Unfortunately, other key properties are also lost: the Rural Hospitals theorem and upper-lattice structure on the set of satisfactory matchings do not hold.

We review prior work in discrete models in the Related Literature section. The model just discussed is even more tightly structured than what had come before, allowing only crowding externalities as opposed to preferences over the identity of one's cohort or even the entire matching. Thus, it will not surprise the reader that those more general models also yield similarly negative results. Thus, so far, no discrete model yields structured solutions and *strategy-proof* rules as in the standard model without externalities.

Related Literature

Matching with Externalities: Our problem is one of matching with externalities. In principle, an agent may have preferences over where all the other agents are matched. Sasaki and Toda (1996); Hafalir (2008); and Bando (2012, 2014) propose and study various notions of stability in such an environment. For the case when there are general externalities *and* several ways for each pair of agents to match—the *matching with contracts* problem—Rostek and Yoder (2020) and Pycia and Yenmez (2023) identify general conditions on preferences under which stability is possible. In most applications, there is a natural structure on possible externalities. For example, agents may care only about who their peers are, not about those matched to other schools/firms (Dutta and Massó 1997; Echenique and Yenmez 2007; Leshno 2021). Narrowing down even further, a large literature considers matching with couples, siblings, or neighbors (Roth 1984; Roth and Peranson 1999; Klaus and Klijn 2005; Kojima, Pathak, and Roth 2013; Ashlagi, Braverman, and Hassidim 2014; Nguyen and Vohra 2018; Dur and Wiseman 2019; Dur, Morrill, and Phan 2022). These studies find that externalities eliminate the structures that have been found in the classical literature. Preference over crowding introduces yet more structure to the problem, but that alone is not sufficient to yield all the standard results from the classical problem.

Tierney (2019) also studies crowding but in an environmental resource allocation problem. In principle, we could treat school resources like environmental

¹³The usual *no justified envy* embodies a “replacement” principle: if i has justified envy of j , then i can simply take the place of j . Copland’s solution is subtly different—one student “joins” another school without replacing anyone else.

resources; however, adapting their equilibrium to our model is not appropriate for several reasons, chief among them being that the concept assumes agents can be allotted arbitrarily small quantities. This is unacceptable for school choice, as it implies that a single school can accommodate all the students in the entire system. Additionally, their equilibrium concept also satisfies *anonymity*. Here, respecting priorities necessitates retaining student identities.

Bogomolnaia and Moulin (2023) study a crowding environment with discrete congestion levels but with probabilistic outcomes. They study envy-freeness and welfare lower bounds in this context, absent any priority order on students. They also find a Walrasian-type solution satisfying these properties, which works even when agents impose different levels of congestion externality on each other.

Walrasian Approaches: Our RCE follows, in spirit, price-based equilibrium notions in matching with the possibility of monetary transfers. Note, however, that the literature in this subsection does not cover externalities. Shapley and Shubik (1971) first proved the existence and structural properties of the core and competitive equilibrium allocations for these models. These properties imply that, for each side of the market, there is a unique undominated utility vector it can achieve in the core. Mechanisms that realize one of these vectors are *efficient* for all and *strategy-proof* (Demange and Gale 1985) for the side of the market whose utility is maximized. In matching models where prices are not fully flexible, e.g., restricted between price floors and ceilings, Drèze Equilibrium and Rationing Price Equilibrium were proposed as alternative notions (Drèze 1975; Talman and Yang 2008; Andersson and Svensson 2014; Herings 2018). Under some mild domain restrictions, there are *constrained efficient* and *group strategy-proof* rules (Andersson and Svensson 2014).

In several matching models without continuous transfers, the usual fairness properties are naturally expressed with an endogenously determined *cutoff* vector (Balinski and Sönmez 1999; Sönmez and Ünver 2010; Abdulkadiroğlu, Che, and Yasuda 2015; Azevedo and Leshno 2016; Dur and Morrill 2018; Leshno and Lo 2021). In the case of school choice, the vector specifies for each school the lowest-priority student able to attend. This is thus a competitive approach—cutoffs determine budget sets for agents, agents maximize therein, and markets clear. Note that the cutoffs in these models play a different role to that of our resource vector. The former control access to schools, while the latter modulates the character of what students eventually consume.

Organization

In Section I, we define the school choice with crowding problem. In Section II, we define the rationing crowding equilibrium. Section III shows that RCEs exist, and Section IV investigates some properties of them, like the Rural Hospitals theorem and the welfare upper lattice. In Section V, we identify a *fair* and *strategy-proof* mechanism, and in Section VI, we find an algorithm that implements this mechanism on a restricted domain of preferences. In Section VII, we discuss the robustness of our results and some applications beyond school choice. Section VIII concludes with open questions.

I. Model

Let S be the finite set of *schools* and N be the finite set of *students*. Each school $s \in S$ has resources in the form of teachers, buildings, money, etc. We aggregate this to a single measure and for each school normalize this to one. When a student attends a school, the number of other students and the policy of the school together determine what fraction of those resources, that is, what *resource ratio*, they consume. Formally, each student's consumption space is $[0, 1] \times S$, where the first component is a resource ratio ρ_s and the second is the school s that they attend. This abstraction of the consumption space drives our theoretical innovations. For some applications, it may be essential that the resource ratio be restricted to *simple* fractions of the form $1/k$. This would be the case when crowding really depends *only* on the number of students at a school. Our model is still useful in these cases since we shall recommend, for each school, a resource ratio that is within one student of being the simple fraction implied by the cohort they admit (see Remark 1). We do not have a formal “outside option” in our model, but we show in online Appendix D that there is a natural embedding of models with outside options into ours; all results remain.

Each student $i \in N$ has a complete, transitive, and continuous *preference relation* R_i over $[0, 1] \times S$. Denote by P_i and I_i the corresponding strict and indifference relations, respectively. We assume that each is monotonic in the resource ratio: for each $\rho_s, \rho'_s \in [0, 1]$ with $\rho'_s > \rho_s$, and each $s \in S$, $(\rho'_s, s) P_i (\rho_s, s)$. Let $\mathcal{R}$ be the set of monotonic preference relations and $\mathbf{R} = (R_i)_{i \in N} \in \mathcal{R}^N$ denote a profile of preferences for students.

Each school $s \in S$ has a maximum *capacity* that is either a positive natural number or infinity. Let $b_s \in [0, 1]$ be such that b_s^{-1} is equal to this capacity, where we take $b_s = 0$ when capacity is infinite; we can then interpret b_s as a lower bound on the resource ratio it can provide. Let $\mathbf{b} = (b_s)_{s \in S}$ denote the profile of bounds for schools. We assume that $\sum_{s \in S} b_s^{-1} \geq |N|$, so all students can be admitted to some school. Each school $s \in S$ has a *priority order* $\succ_s$ over the set of students, where $i \succ_s j$ indicates that i has higher priority than j at s . Let $\succ = (\succ_s)_{s \in S}$ denote the profile of priorities for schools.

A *school choice problem with crowding* (hereafter, just a *problem*) is a tuple $(S, N, \mathbf{R}, \mathbf{b}, \succ)$. The canonical school choice problem of Abdulkadiroğlu and Sönmez (2003) is a special case (see Section IIA for details). A *distribution* is a vector $\rho \in [0, 1]^S$ of resource ratios. A *matching* $\sigma : N \rightarrow S$ places each student at a school. Let $\mathcal{M}$ be the set of matchings, and for each $\sigma \in \mathcal{M}$, write $\sigma[s]$ as the set of students matched to s at σ .¹⁴ An *allocation* is a pair $(\rho, \sigma) \in [0, 1]^S \times \mathcal{M}$ such that for each $s \in S$,

(i) (Distribution Feasibility) $\rho_s \cdot |\sigma[s]| \leq 1$ and

(ii) (Respects Capacity) $\rho_s \geq b_s$.

¹⁴ Generally, for any function f and any element or set y , we let $f[y]$ denote the pre-image of y under f .

Given a distribution $\mathbf{p} \in [0, 1]^S$ and a school $s \in S$, we also write $(\mathbf{p}, s)$ to indicate bundle (ρ_s, s) . Thus, for an allocation $(\mathbf{p}, \sigma)$, each $i \in N$ receives $(\mathbf{p}, \sigma(i))$. We refer to this as *i's component* when the allocation at hand is clear.

A *mechanism* φ recommends, for each preference profile $\mathbf{R}$ in a domain $\mathcal{D} \subseteq \mathcal{R}^N$, an allocation. Denote by $\varphi_i(\mathbf{R})$ student i 's component of the allocation recommended at profile $\mathbf{R}$. We formalize some senses in which these recommendations might either be good or achievable.

We consider fairness among the students. A student should be able to attend any school that is not yet at capacity. If a school is at capacity, then the student may be turned away if all others have higher priority. This embodies the classic *no-envy* condition of Foley (1966) as well as the standard notion of *no justified envy* in school choice. We combine these two ideas formally. Fix a preference profile $\mathbf{R}$. The allocation $(\mathbf{p}, \sigma)$ is *fair* if $(\mathbf{p}, s) P_i(\mathbf{p}, \sigma(i))$ implies that $\rho_s = b_s$ (s is at capacity) and, for each $j \in \sigma[s]$, $j \succ_s i$. We repeat this terminology for the associated property of mechanisms.

Fairness: For each preference profile, the allocation recommended by φ is fair.

There may be several *fair* allocations, and we distinguish those most preferred by the students. Allocation $(\mathbf{p}, \sigma)$ *Pareto-dominates* $(\mathbf{p}', \sigma')$ if for each $i \in N$, $(\mathbf{p}, \sigma(i)) R_i(\mathbf{p}', \sigma'(i))$, and for some $j \in N$, $(\mathbf{p}, \sigma(j)) P_j(\mathbf{p}', \sigma'(j))$. A *student-optimal fair* allocation is *fair* and not Pareto-dominated by any other *fair* allocation.¹⁵

Student-Optimal Fairness: For each preference profile, the allocation recommended by φ is student-optimal *fair*.

Next, we consider the direct revelation incentive compatibility condition.

Strategy-Proofness: For each preference profile $\mathbf{R} \in \mathcal{D}$, each $i \in N$, and each preference relation $R'_i \in \mathcal{R}$ such that $(R'_i, \mathbf{R}_{-i}) \in \mathcal{D}$,

$$\varphi_i(\mathbf{R}) R_i \varphi_i(R'_i, \mathbf{R}_{-i}).$$

One might worry about the potential for a single agent i to report R'_i and induce $(R'_i, \mathbf{R}_{-i}) \notin \mathcal{D}$. However, in all of the problems that we study, this event is negligible, either topologically or probabilistically. We provide a more detailed discussion in Section III.

II. Solution Concept

Our solution concept is fundamentally Walrasian in spirit. There is a publicly announced vector (a distribution) that induces, for each student, a menu of options

¹⁵There may be allocations that are *efficient* (Pareto-undominated by any allocation) but not *fair*. As we are in the context of school choice, we focus on respect for priorities.

(a list of ratio-school pairs), and students select their most preferred. Two main points, though, distinguish our concept from price equilibrium. If too many students select one school, then rationing occurs as opposed to a price increase. Exogenously given priority information ($\succ$) determines which students are matched. Next, once a distribution is announced, they must consume the school at that distribution quantity (i.e., resource-student ratio). They cannot “purchase” more or less resource.

Formally, a *rationing crowding equilibrium* (RCE) is an allocation (ρ, σ) that satisfies three conditions:

1. (Fairness) (ρ, σ) is *fair*.
2. (Exhaustiveness) For each school s with $\sigma[s] \neq \emptyset$,

$$\lfloor \rho_s^{-1} \rfloor = |\sigma[s]|.$$

3. (Inferior Empty Schools) For each school s with $\sigma[s] = \emptyset$, $\rho_s = 1$, and for each $i \in N$,

$$(\rho, \sigma(i)) P_i(\rho, s).$$

Note that our notion begins by offering each student the menu $\{(\rho, s) : s \in S\}$. If $\rho_s > b_s$, then “demand” for s is just as one expects. Anticipating, however, that ρ_s cannot be reduced if it equals b_s , “demand” is first rationed via priorities in this case. Thus, *fairness* is the analog to consumer maximization. The second condition is the key to adapting our notion to the crowding environment and operationalizes the interpretation of ρ_s as a resource ratio. The amount of resources the school provides to each student is, up to rounding error, the total amount of resource (one) divided by the number of students matched to the school. Together with the definition of an allocation (feasibility), this is the analog of market clearing. The third condition states that each student finds any empty school strictly worse than their component of the allocation, and is therefore the analog of the requirement that, at equilibrium, unconsumed commodities should be available for free. Section IIB relates RCE to other notions in the literature encompassing competitive equilibria with price rigidities.

EXAMPLE 2 (An RCE): Let $S = \{s_1, s_2\}$ and $N = \{1, 2, 3\}$. Students’ preferences are given by the following utility functions:

$$u_1(\rho, s_1) = \frac{3}{22} + \rho_{s_1} \quad \text{and} \quad u_1(\rho, s_2) = \rho_{s_2},$$

$$u_2(\rho, s_1) = \frac{7}{12} + \rho_{s_1} \quad \text{and} \quad u_2(\rho, s_2) = \rho_{s_2},$$

$$u_3(\rho, s_1) = \frac{3}{11} + \rho_{s_1} \quad \text{and} \quad u_3(\rho, s_2) = \rho_{s_2}.$$

Each school s has lower bound $b_s = \frac{1}{2}$ (and thus capacity $b_s^{-1} = 2$). School s_1 has the priority order $1 \succ_{s_1} 2 \succ_{s_1} 3$. School s_2 has the priority order $3 \succ_{s_2} 1 \succ_{s_2} 2$.

Allocation $(\mathbf{p}, \sigma) = ((\rho_{s_1}, \rho_{s_2}), (\sigma(1), \sigma(2), \sigma(3))) = ((1/2, 7/11), (s_1, s_1, s_2))$ is an RCE. Fairness: both students 1 and 2 find their component at least as good as others'.¹⁶ Student 3 prefers both student 1 and student 2's component to her own: $u_3(\mathbf{p}, \sigma(3)) = 7/11 < 3/11 + 1/2 = u_3(\mathbf{p}, s_1)$. Since $\rho_{s_1} = b_{s_1} = 1/2$ and $1 \succ_{s_1} 2 \succ_{s_1} 3$, however, fairness is still satisfied. Exhaustiveness: We have $\lfloor \rho_{s_1}^{-1} \rfloor = |\sigma[s_1]| = 2$ and $\lfloor \rho_{s_2}^{-1} \rfloor = |\sigma[s_2]| = 1$. Inferior empty schools is trivially satisfied, as there is no empty school.

This example is also among the most extreme cases of mismatch between the number of students matched to a school and the reciprocal of the resource ratio. Since there is only one student at s_2 , we should hope that $\rho_{s_2} = 1$. Let $\rho'_{s_2} = \rho_{s_2} + \epsilon$ and consider $\mathbf{p}' = (\rho_{s_1}, \rho'_{s_2})$. For student 1, $u_1(\mathbf{p}', \sigma(1)) = \frac{7}{11} < \frac{7}{11} + \epsilon = u_1(\mathbf{p}', s_2)$. Since $\rho'_{s_2} > b_{s_2} = 1/2$ and student 1 prefers (ρ', s_2) to their own component, we have a violation of fairness.

In this problem, the entire set of RCEs is

$$\left\{ (\mathbf{p}, \sigma) : \rho_{s_1} = \frac{1}{2}, \frac{1}{2} < \rho_{s_2} \leq \frac{7}{11}, \sigma(1) = s_1, \sigma(2) = s_1, \sigma(3) = s_2 \right\} \\ \cup \left\{ (\mathbf{p}', \sigma') : \rho'_{s_1} = \frac{1}{2}, \frac{7}{11} \leq \rho'_{s_2} \leq \frac{17}{22}, \sigma'(1) = s_2, \sigma'(2) = s_1, \sigma'(3) = s_1 \right\}.$$

Note that we can have $\rho_{s_2} > 7/11$ when the matching is changed but that in all cases, school s_2 will have only one student, and $\rho_{s_2} < 1$.

REMARK 1 (Discrepancy between ρ_s and a Simple Ratio of the Form $1/|\sigma[s]|$): The magnitude of the discrepancy at any RCE is

$$0 \leq \frac{1}{|\sigma[s]|} - \rho_s = \frac{1}{|\sigma[s]|} - \frac{1}{\rho_s^{-1}} < \frac{1}{|\sigma[s]|} - \frac{1}{\lfloor \rho_s^{-1} \rfloor + 1} = \frac{1}{|\sigma[s]|} - \frac{1}{|\sigma[s]| + 1}.$$

Thus, the difference between our modeled ratio and a strictly interpreted resource-student ratio is at most the addition of one more student. This is a negligible difference in most applications.

A. Connection to the Standard School Choice Model

Consider the canonical school choice model of Abdulkadiroğlu and Sönmez (2003). We show how to embed this problem into school choice with crowding, then relate solution concepts across models.

Let schools S , students N , and priorities $\succ$ be defined as before. For each student $i \in N$, let P_i^* be a strict preference relation over schools and $\mathbf{P}^* = (P_i^*)_{i \in N}$ be the profile of such preferences. For each school $s \in S$, let $c_s^* \in \mathbb{N}$ be its capacity

¹⁶ $u_1(\mathbf{p}, \sigma(1)) = \frac{3}{22} + \frac{1}{2} = \frac{7}{11} = u_1(\mathbf{p}, s_2)$, and $u_2(\mathbf{p}, \sigma(2)) = \frac{7}{12} + \frac{1}{2} > \frac{7}{11} = u_2(\mathbf{p}, s_2)$.

and $\mathbf{c}^* = (c_s^*)_{s \in S}$ be the capacity profile for S . A *school choice problem* is a tuple $(S, N, \mathbf{P}^*, \mathbf{c}^*, \succ)$. An allocation is a matching $\sigma : N \rightarrow S$ such that for each $s \in S$, $|\sigma[s]| \leq c_s^*$. We recall two central properties in this model. A matching σ satisfies *no justified envy* if for each $i \in N$, there is no $j \in N \setminus \{i\}$ such that $\sigma(j) P_i^* \sigma(i)$ and $i \succ_{\sigma(j)} j$. A matching σ satisfies *nonwastefulness* if for each $i \in N$, and $s \in S$, $s P_i^* \sigma(i)$ implies $|\sigma[s]| = c_s^*$.

We now construct an associated school choice problem with crowding $(S, N, \mathbf{R}, \mathbf{b}, \succ)$. Let S , N , and $\succ$ be as in the school choice problem; only $\mathbf{R}$ and $\mathbf{b}$ need adjustment. For each $i \in N$, let R_i be such that for each $s, s' \in S$, $s P_i^* s'$ if and only if $(0, s) P_i(1, s')$. That is, at any distribution level, i prefers s to s' in R_i . For each $s \in S$, let $b_s = \frac{1}{c_s^*}$. Thus, the classical model embeds in ours as a preference restriction. The externality is still present: students are worse off when they have more classmates. However, on the restricted domain of classical preferences, there is no way for students to reveal this fact through their choices.

If we wish, we may include a special school, ϕ , in our model with $b_\phi < 1/|N|$. When the canonical school choice model is embedded in ours, this school may function as an outside option in the standard sense since it can accept all students. This is despite the fact that in our model, we have elected not to consider the outside option.

PROPOSITION 1: *Fix a school choice problem. The following statements are equivalent:*

- (i) σ satisfies no justified envy and nonwastefulness, and
- (ii) There is a distribution $\mathbf{p}$ such that $(\mathbf{p}, \sigma)$ is an RCE for the associated school choice problem with crowding.

PROOF:

Fix a school choice problem $(S, N, \mathbf{P}^*, \mathbf{c}^*, \succ)$. Let $(S, N, \mathbf{R}, \mathbf{b}, \succ)$ be an associated school choice problem with crowding. Note that, as an implication of our construction, for each $i \in N$, each $s \in S$, and each $\mathbf{p} \in [0, 1]^S$, we have $s P_i^* \sigma(i)$ if and only if $(\mathbf{p}, s) P_i(\mathbf{p}, \sigma(i))$.

(i) $\Rightarrow$ (ii): Let σ be a matching satisfying the two properties. Let $\mathbf{p} \in [0, 1]^S$ be such that for each empty school $s \in S$, $\rho_s = 1$ and for each nonempty school $s \in S$, $\rho_s^{-1} = |\sigma[s]|$. Thus, $(\mathbf{p}, \sigma)$ is an allocation and exhaustive. We show the other two properties of RCE one school at a time. Let $s \in S$ be such that $\sigma[s] = \emptyset$. By *nonwastefulness*, for each $i \in N$, $\sigma(i) P_i^* s$, and so $(\mathbf{p}, \sigma(i)) P_i(\mathbf{p}, s)$. The latter vacuously implies that *fairness* is satisfied for s . By construction, $\rho_s = 1$, and so *inferior empty schools* is satisfied for s . Let $s \in S$ be such that $\sigma[s] \neq \emptyset$. *Inferior empty schools* is vacuously satisfied. Let $i \in N$ be such that $s P_i^* \sigma(i)$. By construction and *nonwastefulness*, $\rho_s^{-1} = |\sigma[s]| = c_s^* = b_s^{-1}$. By *no justified envy*, for each $j \in \sigma[s]$, $j \succ_s i$. Combining these two, we have that *fairness* is satisfied for s .

(ii) $\Rightarrow$ (i): Let $(\mathbf{p}, \sigma)$ be an RCE, $i \in N$, and $s \in S$ be such that $(\mathbf{p}, s) P_i(\mathbf{p}, \sigma(i))$, and so $s P_i^* \sigma(i)$. *Inferior empty schools* implies that $\sigma[s] \neq \emptyset$. By *fairness*, for each $j \in \sigma[s]$, $j \succ_s i$. So *no justified envy* is satisfied at s . By exhaustiveness, *fairness*, and construction, $|\sigma[s]| = \lfloor \rho_s^{-1} \rfloor = b_s^{-1} = c_s^*$, and so σ is *nonwasteful* at s . ■

B. Competitive Foundations of RCE

Our RCE concept is related to notions of competitive equilibria when prices exhibit rigidities. In the classical exchange problem, a price ceiling may cause demand to outstrip supply, resulting in the failure of market clearing and thus nonexistence of equilibria. Drèze (1975) proposed and showed existence for a notion where prices are constrained by ceilings or floors and rationing occurs at such boundaries. In addition to prices, the notion includes a rationing scheme that specifies limits for the net trades of agents. Likewise, in matching models with price controls, Talman and Yang (2008); Andersson and Svensson (2014); and Herings (2018) introduce similar Drèze-style equilibrium concepts. None of these models consider consumption externalities. Our RCE can be seen as a conceptual parallel to their notions, but for the environment where agents have preferences over crowding.

To explore this relationship, we define several Drèze-style equilibrium notions for a school choice with crowding problem $(S, N, \mathbf{R}, \mathbf{b}, \succ)$, then compare them to those above. A rationing scheme provides each student with a set of schools that is available for them to consume. For each student $i \in N$, let $Q_i \subseteq S$ be the set of schools available to them. Let $\mathbf{Q} = (Q_i)_{i \in N}$ be the rationing scheme. A special case (defined below) is when rationing relies on priorities associated with the schools.

The *rationed demand set* of student i at distribution $\mathbf{p}$ and rationing scheme $\mathbf{Q}$ is

$$D_i(\mathbf{p}, Q_i) = \{s \in Q_i : (\mathbf{p}, s) R_i(\mathbf{p}, s'), \forall s' \in Q_i\}.$$

Consider a tuple $(\mathbf{p}, \sigma, \mathbf{Q})$ consisting of an allocation and a rationing scheme. Each equilibrium notion is defined by subsets of the conditions below:

1. Each student is matched to a school in their rationed demand set.
2. For each school s with $\sigma[s] \neq \emptyset$, $\lfloor \rho_s^{-1} \rfloor = |\sigma[s]|$.
3. For each pair of student i and school s such that $s \notin Q_i$, $(\mathbf{p}, s) P_i(\mathbf{p}, \sigma(i))$ and $\rho_s = b_s$.
4. For each pair $i, j \in N$, and school $s \in S$ such that $s \notin Q_i$ and $j \in \sigma[s]$, $j \succ_s i$.

A *crowding Drèze equilibrium* (CDE) is a tuple $(\mathbf{p}, \sigma, \mathbf{Q})$ that satisfies conditions 1–3. Note that $\mathbf{Q}$ is a general rationing scheme and does not depend on $\succ$. When $\mathbf{Q}$ is consistent with the priority profile $\succ$ (condition 4), then we say that it is a *priority-compatible Drèze equilibrium* (PCDE). Our RCE is a further refinement of PCDE by additionally imposing the *inferior empty schools* condition.

Without crowding, these notions coincide with several of those in the literature. Condition 2 is the key difference: the number of students matched to a school is nearly the distribution associated with the school. Without this coupling, we revert back to the interpretation of the distribution as the price. Without condition 2, CDE is the notion in Talman and Yang (2008), and if priorities are used, then PCDE, with additional requirement of *constrained efficiency*, is that in Andersson and Svensson

(2014).¹⁷ Finally, notice that when students' preferences satisfy the standard monotonicity and continuity assumptions, the existence for each of the aforementioned equilibrium notions is guaranteed. This is not true when there is crowding, as Example 3 demonstrates. With a mild domain restriction, however, we can recover existence (Theorem 1).

III. Existence of RCEs

There are profiles in $\mathcal{R}^N$ that do not admit an RCE. Such profiles are rare; profiles that *do* admit an RCE are generic. We first present an example and then introduce a domain restriction that excludes these cases and ensures existence.

EXAMPLE 3 (The Nonexistence of RCE): Let $S = \{s_1, s_2, s_3\}$ and $N = \{1, 2, 3\}$. Students have the following utility functions:

$$\begin{aligned} u_1(\mathbf{p}, s_1) &= \rho_{s_1}, & u_1(\mathbf{p}, s_2) &= \rho_{s_2}, & u_1(\mathbf{p}, s_3) &= -\frac{1}{2} + \rho_{s_3}, \\ u_2(\mathbf{p}, s_1) &= \rho_{s_1}, & u_2(\mathbf{p}, s_2) &= \rho_{s_2}, & u_2(\mathbf{p}, s_3) &= -\frac{2}{3} + \rho_{s_3}, \\ u_3(\mathbf{p}, s_1) &= \rho_{s_1}, & u_3(\mathbf{p}, s_2) &= \rho_{s_2}, & u_3(\mathbf{p}, s_3) &= -\frac{3}{4} + \rho_{s_3}. \end{aligned}$$

Each school is allowed to have any priority order, and $b_{s_1} = b_{s_2} = 1/2$, and $b_{s_3} = 0$.

We show that there is no RCE. By contradiction, suppose that there is. First we claim that no student is matched with school s_3 . If there were, then by either exhaustiveness or inferior empty schools, at least one of the other two schools, i.e., s_1 and s_2 , should have a ratio greater than $1/2$. Thus, the student matched with s_3 prefers the school with a ratio greater than $1/2$, contradicting **fairness**.

Since $b_{s_1} = b_{s_2} = 1/2$, in the case where s_1 takes two students, the ratio at s_1 is $1/2$ while s_2 only takes one student, and by exhaustiveness, has a ratio greater than $1/2$. Thus, any student matched with s_1 prefers s_2 , contradicting **fairness**. The same reasoning works for the case where s_2 takes two students.

We now introduce our domain restriction. Given a preference profile $R \in \mathcal{R}^N$, two schools s_1 and s_{k+1} are *connected by indifference* if there is a distribution $\mathbf{p}$, a sequence of distinct students $\{i_1, \dots, i_k\}$, and a sequence of schools (at least two distinct) $\{s_1, \dots, s_{k+1}\}$ such that

- (1) $\rho_{s_1} = \frac{1}{n}$ and $\rho_{s_{k+1}} = \frac{1}{m}$ for some $m, n \in \{1, \dots, |N|\}$ and
- (2) $(\mathbf{p}, s_\ell) I_{i_\ell}(\mathbf{p}, s_{\ell+1})$ for each student i_ℓ with $1 \leq \ell \leq k$.

¹⁷They also require the standard Walrasian requirement that unassigned objects have their reserve price, which in our model corresponds to $\rho_s = 1$. In Section VIIA we explain the consequences of replacing *inferior empty schools* with this requirement.

Note that $\mathbf{p}$ in this case need not be part of an RCE, nor even be compatible with any feasible allocation; it is an arbitrary vector. A preference profile is connected by indifference if there are two schools that are connected by indifference. A preference domain satisfies *no connection by indifference* (NCBI) if it contains no preference profile that is connected by indifference. Denote by $\mathcal{D} \subseteq \mathcal{R}^N$ the subdomain of *all* profiles that are not connected by indifference.¹⁸ Call this *the* NCBI domain. The preference profile presented in Example 3 is not in the NCBI domain, as all three students are indifferent between $(1/2, s_1)$ and $(1/2, s_2)$. On the other hand, the preference profile presented in Example 2 is not connected by indifference.

The NCBI domain $\mathcal{D}$ is not rectangular but is open and dense in the full domain. Thus, we may invoke the Kuratowski-Ulam theorem to say that for a generic profile of the others, $\mathbf{R}_{-i}$, and a generic manipulation R'_i of i , the resulting profile $(R'_i, \mathbf{R}_{-i}) \in \mathcal{D}$.¹⁹ For our algorithm, however, we consider a natural finite-dimensional preference domain. Restricting to this domain, it is easy to see that the Lebesgue measure of the event that an agent forces the problem outside the domain is zero (see Remark 2 in Section VI).

THEOREM 1: *Each preference profile in the NCBI domain admits an RCE.*

The proof of Theorem 1 is in online Appendix A. We provide a sketch of the argument here. Start from a *fair* allocation. That is, we allow for the discrepancy between ρ_s^{-1} and the actual number of matched students to be arbitrarily large. The existence of *fair* allocations is trivial: set the distribution equal to vector $\mathbf{b}$ and run the deferred acceptance algorithm.

The set of distributions that generate RCE, if nonempty, lie in the upper envelope of the set of distributions that generate a *fair* allocation. So, given that a *fair* allocation always exists, our argument begins with one of these and seeks to increase the distribution vector while maintaining *fairness*. We do this with some graph theoretic tools.

With a distribution fixed, we study a graph with vertices S and such that an arc represents a student at a given school who finds another school at least as good. A *source set* is a set of vertices such that no arcs enter the set (there may be arcs among vertices in the set, so that no vertex is a source on its own). Lemma 1 in online Appendix A.3 tells us that if we can find a set of schools that is a source set in our graph, and for which all schools fail our exhaustiveness condition, then we can perturb upward the ratios for these schools and arrive at another *fair* allocation.

Thus, our goal is to move students among schools such that we do not violate *fairness* and that we find a source set. This is the only part of the argument that requires NCBI, and it is achieved in Lemma 2 in online Appendix A.3. In sum,

¹⁸NCBI is similar to the identically named condition in Andersson and Svensson (2014), although the two are applied to different environments. Our condition is stronger than theirs, and the latter is not sufficient to show existence of RCE in our model.

¹⁹A set is generic if it is the countable intersection of open-and-dense sets. When a particular generic set is fixed (e.g., a subdomain), its elements are called *generic*. Endow $\mathcal{R}$ with the topology of closed convergence (Hildenbrand 2015). Thus, $\mathcal{R}$ is metrizable and second-countable. Assume $\mathcal{D}$ is open and dense, which implies that it is both generic (also called comeager) in $\mathcal{R}^N$ and has the Baire property. Given $\mathbf{R}_{-i} \in \mathcal{R}^{N \setminus \{i\}}$, let $\mathcal{D}_{\mathbf{R}_{-i}} = \{R'_i \in \mathcal{R} : (R'_i, \mathbf{R}_{-i}) \in \mathcal{D}\}$. Then the Kuratowski-Ulam theorem (see Kechris 2012) implies that the set $\{\mathbf{R}_{-i} \in \mathcal{R}^{N \setminus \{i\}} : \mathcal{D}_{\mathbf{R}_{-i}} \text{ is generic in } \mathcal{R}\}$ is generic in $\mathcal{R}^{N \setminus \{i\}}$.

beginning with a *fair* allocation, if it is not an RCE, then we can either increase the ratio of some school or find another *fair* and Pareto-dominating allocation. Along the way, we eliminate the problem of empty schools by simply putting students in them; NCBI ensures they will not hinder us in finding a source set.

Having established that we can perturb upward any *fair* allocation that is not an RCE, it remains only to make a limit argument. This is done in Theorem 6 in online Appendix A.4, which actually proves Theorem 1 and Proposition 3.

It is known that the exact auction of Demange, Gale, and Sotomayor (1986) and its variants crucially depend on the quasi-linear assumption, and when agents have non-quasi-linear preferences, they are not appropriate methods to show the existence of equilibrium (Zhou and Serizawa 2023). Instead, the salary adjustment process of Crawford and Knoer (1981) and Kelso and Crawford (1982) and its variants are frequently used to establish the existence of equilibrium (Herings 2018; Fleiner et al. 2019). In general, this method requires that agents always choose their favorite schools among those who have not rejected them yet, and an agent's welfare is independent of the number of tentatively matched agents. This is not true in our model; thus, the method fails. Another well-known method is via Scarf's Lemma (Quinzii 1984): one first shows existence when agents have piecewise quasi-linear utility functions and then takes the limit, approximating the original continuous utility functions. It is unclear how to establish the existence of RCEs even when agents have piecewise quasi-linear utility functions, and the main challenges still come from showing *exhaustiveness*.

IV. Structural Properties of RCEs

In the classical model, the set of *no justified envy* and *nonwasteful* allocations has a number of remarkable structural properties. Among them, the welfare lattice and the so-called *Rural Hospitals theorem* stand out as particularly important (Roth 1986; Roth and Sotomayor 1990). On the NCBI domain, we find analogous, sometimes identical properties. RCEs form a welfare upper lattice, which further implies the classical upper lattice result via the embedding. We also show that the number of students matched to any school remains constant across all RCEs, which establishes a *Rural Hospitals theorem* for our environment.

To help prove these results, we find a decomposition result that also has analogs in earlier literature but was previously unknown. We require, first, a definition. A sequence of distinct students $(i_1, \dots, i_k)$, $k \geq 2$, constitutes a *trading cycle* from a matching σ to a matching τ if $\tau(i_l) = \sigma(i_{l+1})$ for each $1 \leq l \leq k-1$ and $\tau(i_k) = \sigma(i_1)$. If a student i is such that $\sigma(i) = \tau(i)$, then we say that they *self cycle* from σ to τ . A *cycle decomposition* from σ to τ is a collection of trading cycles and self cycles where (i) for each $i \in N$ with $\sigma(i) \neq \tau(i)$, i is in a trading cycle; (ii) for each $i \in N$ with $\sigma(i) = \tau(i)$, i is in a self cycle; and (iii) trading cycles are disjoint.

PROPOSITION 2: *Consider two arbitrary RCEs, (ρ, σ) and (γ, τ) , for a given preference profile from the NCBI domain:*

- (i) *There is a (not necessarily unique) cycle decomposition from σ to τ .*

- (ii) *Given a cycle decomposition from σ to τ , if student i experiences a strict welfare improvement (welfare reduction) in going from (ρ, σ) to (γ, τ) , then students in the same trading cycle as student i experience a nondecreasing welfare change (nonincreasing welfare change).*

The proof of Proposition 2 is in online Appendix A.2. What follows is a sketch of the argument. Schools whose ratio either increases or remains the same cannot take on more students (*exhaustiveness*). Students attending a school whose ratio *increases* must be better-off, as its increase causes it to rise above its lower bound and thereby be available for all. Then, using Walrasian-type reasoning, we find that these better-off students must go to a school whose ratio must not have decreased. This, of course, “closes off” the set of such schools and yields the proposition for them. The proposition for the other students comes from feasibility, since the schools whose ratios have decreased must take in the rest of the students, and from the properties of RCE—these schools must also have “high enough” ratios by *exhaustiveness*.

The full argument is complicated by several technical details, most difficult among them being what if $(\gamma, \tau(i)) I_i (\rho, \sigma(i))$, $\gamma_{\sigma(i)} = \rho_{\sigma(i)} > b_{\sigma(i)}$, and $\gamma_{\tau(i)} = \rho_{\tau(i)} = b_{\tau(i)}$? Student i may displace j at $\tau(i)$ who has $i \succ_{\tau(i)} j$ since i 's indifference means that j 's presence at $\tau(i)$, under allocation (ρ, σ) , was not a priority violation. Student j 's welfare may drop, and she may envy i at (γ, τ) , without priority violation. The problem is that seemingly nothing prevents i from being part of a trading path along which some previous student has increasing welfare and all other previous students have nondecreasing welfare. For this case, we invoke NCBI, and the earlier Walrasian-type reasoning, as we find that such a student i must actually be the end of an indifference path originating at a school $u \in S$ with $\rho_u = b_u$. Thus, it appears that NCBI is essential for this decomposition to hold.

Note the significance of statement (ii) in the proposition. In general, if $\gamma_s^{-1} > \rho_s^{-1}$, then s might be able to take on more students under (γ, τ) , and so may be the endpoint of a trading *path* rather than a member of a trading cycle. In fact, this occurs among the *fair* allocations, and so our decomposition does not hold on that larger set of allocations.

Proposition 2 is reminiscent of the classical decomposition property of the marriage market (Roth and Sotomayor 1990). Moving from one stable outcome to another, there is a one-to-one correspondence between agents on the one side who have strictly increased welfare (respectively, strictly reduced welfare) and those on the other side whose welfare is strictly reduced (respectively, strictly increased). This property holds for competitive equilibrium/core outcomes in the two-sided matching models with transfers as well (Demange and Gale 1985). Nevertheless, such a decomposition does not hold in our model. Instead, we provide a generalized decomposition by associating welfare changes with trading cycles of components in an allocation. In our language, the existing decomposition results can be read as follows: if an agent experiences a strict welfare improvement, all the other agents in the same trading cycle will also experience strict welfare improvement. In contrast, Proposition 2 admits the possibility of unchanged welfare in a trading cycle.

We state our version of the *Rural Hospitals theorem*—a clear corollary of Proposition 2.

THEOREM 2: *Fix a preference profile from the NCBI domain and let (ρ, σ) and (γ, τ) be two RCEs for this profile. Then for each school $s \in S$, the number of students matched to s under σ is equal to the number of students matched to s under τ .*

Consider the set of RCEs in Example 2. There are two RCE matchings. The first one is $\sigma = (\sigma(1), \sigma(2), \sigma(3)) = (s_1, s_1, s_2)$. The second one is $\sigma' = (\sigma'(1), \sigma'(2), \sigma'(3)) = (s_2, s_1, s_1)$. In either matching, school s_1 is always matched with two students, and school s_2 is matched with one student. The standard Rural Hospitals theorem also states that if a school has unfilled seats at a stable matching, then it matches to the same set of students in every stable matching. Example 2 also demonstrates that this fact is not true under crowding.

Given two distributions, ρ and γ , let $\rho \vee \gamma \in [0, 1]^S$ denote the vector whose s component, for each $s \in S$, is $\max\{\rho_s, \gamma_s\}$.

THEOREM 3: *Assume that (ρ, σ) and (γ, τ) are RCEs for a preference profile from the NCBI domain. There is a matching μ such that $(\rho \vee \gamma, \mu)$ is an RCE, and for each $i \in N$,*

$$(\rho \vee \gamma, \mu(i)) R_i \max_{R_i} \{(\rho, \sigma(i)), (\gamma, \tau(i))\}.$$

The full proof of Theorem 3 is in online Appendix A.2. Similar to extant results of similar character, a decomposition result, in our case Proposition 2, is the main tool. We simply begin with one of the two RCEs, say (ρ, σ) , and to arrive at a candidate matching, μ , execute all the welfare-non-decreasing trading cycles from σ to τ . Then we show that $(\rho \vee \gamma, \mu)$ is an RCE. If $(\rho \vee \gamma)_s = \gamma_s > \rho_s$, then any student $i \in \sigma[s]$ must have increased welfare, as otherwise, they would prefer s at (γ, τ) , and the previous inequality gives $\gamma_s > b_s$. Then Proposition 2 and some supporting results in the online Appendix allow us to conclude that i is part of a cycle among schools whose resource ratio is at least as high under γ as it is under ρ . This further allows us to use the feasibility of (γ, τ) to conclude the feasibility of $(\rho \vee \gamma, \mu)$. Since all students are partitioned by Proposition 2, the execution of these cycles will not interfere with each other. The foregoing argument studied the case when $\rho \neq \gamma$. However, the decomposition holds equally well when $\rho = \gamma$ and so demonstrates that RCEs induce an upper lattice in welfare space.

It is in general not true that the existence of an upper lattice in distributions implies the existence of an upper lattice in welfare (see Example 4). Here again, the NCBI domain seems crucial.

Since we have an upper lattice in welfare space, a limit argument is sufficient to show the existence of a greatest RCE welfare vector. Several RCEs may induce this vector, all of which have the same distribution. Any RCE that induces this vector is called *maximal*. For Example 2, one such maximal RCE is given by distribution $(1/2, 17/22)$ paired with matching (s_2, s_1, s_1) . We formalize the foregoing observations as follows.

PROPOSITION 3: *Given a preference profile R from the NCBI domain,*

- (i) *there is a greatest RCE distribution $\rho^*(R)$,*
- (ii) *there is a maximal RCE, with distribution $\rho^*(R)$, and all students are indifferent between all maximal RCEs, and*
- (iii) *among all RCEs, only the maximal RCEs satisfy student-optimal fairness.*

The proof of Proposition 3 is in online Appendix A.4. It is not true that an RCE compatible with $\rho^*(R)$ always maximizes students' welfare; maximal RCEs are a strict subset of the RCEs that are compatible with $\rho^*(R)$.

We conclude this section with two remarks. First, in Section IIA we showed that the standard school choice model can be embedded in our model. It follows that the standard Rural Hospitals theorem of Roth (1986) is also a corollary of Theorem 2.²⁰ Theorem 3 and Proposition 3 mirror the standard welfare lattice results in Roth and Sotomayor (1990), but here, we only consider the student side of the market and its corresponding direction in the lattice. Second, even when RCEs exist for profiles outside the NCBI domain, the structural results above may fail to hold.

EXAMPLE 4 (Lack of Structure on the General Domain): *The reader will observe that the structures above fail for the same reason they do in the standard school choice model when students' preferences may have indifferences.*

Let $S = \{s_1, s_2, s_3, s_4\}$ and $N = \{1, 2, 3\}$. Students have the following preferences. For each $\rho \in [0, 1]^S$,

$$(\rho, s_1) I_1(\rho, s_2) P_1(\rho, s_3) P_1(\rho, s_4)$$

$$(\rho, s_1) P_2(\rho, s_4) P_2(\rho, s_3) I_2(\rho, s_2)$$

$$(\rho, s_2) P_3(\rho, s_3) P_3(\rho, s_1) I_3(\rho, s_4).$$

Each school has unit capacity, i.e., $b_{s_1} = b_{s_2} = b_{s_3} = b_{s_4} = 1$. Schools have the following priority orders: $1 \succ_{s_1} 2 \succ_{s_1} 3$; $1 \succ_{s_2} 3 \succ_{s_2} 2$; $3 \succ_{s_3} 2 \succ_{s_3} 1$; and $2 \succ_{s_4} 3 \succ_{s_4} 1$.

Let $\rho = (1, 1, 1, 1)$. There are two RCEs compatible with ρ , clearly making $\rho^*(R) = \rho$. The first one is (ρ, σ) such that $(\sigma(1), \sigma(2), \sigma(3)) = (s_1, s_4, s_2)$. The second one is (ρ, τ) such that $(\tau(1), \tau(2), \tau(3)) = (s_2, s_1, s_3)$. It is not hard to see that there is no trading cycle from σ to τ , and so Proposition 2 fails to hold. Also at (ρ, σ) , school s_3 is empty, and at (ρ, τ) , school s_4 is empty. Therefore, Theorem 2 fails to hold. Note that in both of the RCEs mentioned, two students get their favorite possible bundle, and one gets their second favorite. Thus, the only way to improve upon this is with matching $(\mu(1), \mu(2), \mu(3)) = (s_1, s_1, s_2)$ or matching

²⁰ Recall that student preferences in the embedding are extreme and induce strict preferences over schools, independent of resource levels. By *exhaustiveness* and Theorem 2, if a school s is unfilled ($\rho_s > b_s$) at some RCE, then it remains so for all RCEs. If a student is moved out of an unfilled school, then by *fairness*, this student is better-off at the new RCE. By Proposition 2, they are part of a trading cycle in which all students are better-off. This means the student who enters s previously preferred s , contradicting *fairness*.

$(\mu'(1), \mu'(2), \mu'(3)) = (s_2, s_1, s_2)$, both of which are infeasible. However, student 2 prefers allocation (ρ, τ) over (ρ, σ) , whereas student 3 prefers (ρ, σ) . Thus, Theorem 3 fails to hold. Since there are no maximal RCEs, Proposition 3 fails to hold as well.

Examples 3 and 4 highlight the role of our domain restriction, NCBI. However, it is worth noting that the structural properties hold under a weaker restriction. In particular, we may weaken the first condition in the definition of connected by indifference. However, the weakened domain restriction is *not* sufficient for our proof of the existence of RCE. See online Appendix A.3 for details.

V. Maximal RCE Mechanisms

Fixing an environment of schools, students, and lower bounds, for each profile in the NCBI domain, there is a nonempty set of maximal RCE allocations, between which all students are indifferent (Proposition 3). A maximal RCE mechanism is a function that selects, for each profile in the NCBI domain, a maximal RCE allocation; we do not define these mechanisms on the full domain. Thus, all maximal RCE mechanisms are welfare equivalent.

THEOREM 4: *Any maximal RCE mechanism on the NCBI domain is strategy-proof.*

The full proof of Theorem 4 is in online Appendix B. A student who wishes to manipulate the rule must consider their influence on both the match and the greatest RCE distribution. To understand why a manipulation will fail, we first introduce a key property of this object, called *connectedness* (see Lemma 3 in online Appendix B). Consider a student i who is not weakly envied by another student. That is to say, all students not attending i 's school strictly prefer their outcome to i 's. Then it is not feasible to raise the resource ratio at i 's school; if it were, we could do so by a small amount, make i and her classmates happier, and not induce any envy from other students. Thus, for each student i , one of the following is true: (i) some student j from another school finds i 's outcome at least as good as her own or (ii) the resource at i 's school is *totally exhausted*. It follows that, for each student for whom (i) is true, we can find a sequence of students $\{j, \dots, k, l\}$ such that l finds i 's outcome at least as good as her own, k finds l 's outcome at least as good as her own, etc., and j 's school exhausts its resource. This is the so-called *connectedness* property, also seen in minimal-price Walrasian rules for allocating objects and money (Morimoto and Serizawa 2015). We shall call the sequences that define these connections *weak-envy paths*.

Suppose a student manipulates a problem and is then matched to school s . If the student were to cause the resource level at s to increase, then, of course, it cannot be at its lower bound and therefore, under RCE, is available to any student who prefers it. Since there is a weak-envy path leading from an exhausted school to s , the welfare of some students on the path leading into s must be increased. This is done either by raising the resource ratio at their schools or by moving them. We observe that the raising of resource ratios must propagate up the path, leading to an infeasible

outcome, as the head of the path is a school that has exhausted its resource. More generally, such arguments lead us to the conclusion that agents cannot manipulate by altering the distribution. However, we then show that a manipulation in our model given a fixed distribution can be mapped into a manipulation of the student-optimal stable match in the classical school choice model. This is, of course, impossible.

VI. An Algorithm for Maximal RCE

As discussed after Theorem 1, it is nontrivial to find an algorithm that calculates an RCE in a finite number of steps, in particular, when students have general preferences, as studied here. We can, however, on a restricted domain. Say a preference relation $R_i \in \mathcal{R}$ is linear if there is a valuation function $v_i: S \rightarrow \mathbb{R}_{++}$ such that for each $(r, s), (r', s') \in [0, 1] \times S$, $(r, s) R_i (r', s')$ if and only if $r v_i(s) \geq r' v_i(s')$. For the remainder of this section, we fix a profile of linear preferences $\mathbf{R} \in \mathcal{D}$.

REMARK 2 (NCBI in Linear Preferences): *In the domain of linear preferences, a violation of NCBI is a sequence $\{1, \dots, m\}$ of students, a sequence $\{1, \dots, m+1\}$ of schools, and a distribution $\rho \in [0, 1]^S$ such that, for each j on the sequence, $\rho_j v_j(j) = \rho_{j+1} v_j(j+1)$, and for the two end points, $\rho_1^{-1}, \rho_{m+1}^{-1} \in \{1, 2, \dots, |N|\}$. By recursion of the foregoing equation on this sequence, we find*

$$\frac{v_m(m+1)}{v_m(m)} = \frac{\rho_1}{\rho_{m+1}} \prod_{k=1}^{m-1} \frac{v_k(k)}{v_k(k+1)}.$$

Note that $v_m(\cdot)$ is preference-equivalent to $v_m(\cdot)/v_m(m)$, and so given the preferences of other students $1, \dots, m-1$, there is exactly one report of student m that completes this particular chain of indifference (given by the equation above). There are finitely many beginning and end points of a possible chain, and for each of these pairs, finitely many ways to arrange the preferences of the $N \setminus \{m\}$ students to form a chain. Thus, student m has finitely many reports to induce a failure of NCBI, and so, invoking the Fubini-Tonelli theorem, students reporting linear preferences almost surely report a profile that is in $\mathcal{D}$.

The algorithm we introduce uses tools from the multi-item auction (Demange, Gale, and Sotomayor 1986) and the Deferred Acceptance algorithm (Gale and Shapley 1962), presented in the framework of Drèze equilibrium (Section IIB). In brief, starting from an initialized distribution, the algorithm will decrement it continuously, pausing occasionally to elicit demands and “reject” some students from some schools; the rejected students cannot thereafter demand these schools. In such cases, time is advanced discretely, and, if possible, decrementing is re-initialized.

We start with some useful definitions. Let $z \in \mathbb{R}$ be a moment in time and $\rho_z = (\rho_{sz})_{s \in S}$ be an associated distribution. At z , each student $i \in N$ has a set $Q_{iz} \subseteq S$ of schools available to them, and so their rationed demand set is $D_i(\rho_z, Q_{iz}) \subseteq Q_{iz}$, shortened to D_{iz} for simplicity. Let $\mathbf{Q}_z = (Q_{iz})_{i \in N}$ be the rationing scheme at time z . A set $S' \subseteq S$ is *in excess demand* at $(\rho_z, \mathbf{Q}_z)$ if

$$(1) \quad |\{i \in N : D_{iz} \subseteq S'\}| > \sum_{s \in S'} \lfloor \rho_s^{-1} \rfloor, \text{ and}$$

(2) for each nonempty $S'' \subsetneq S'$,

$$|\{i \in N : D_{iz} \subseteq S' \text{ and } D_{iz} \cap S'' \neq \emptyset\}| > \sum_{s' \in S''} \lfloor \rho_s^{-1} \rfloor.$$

Recall that $\lfloor \rho_s^{-1} \rfloor$ is the maximum number of students that a school can accept if they distribute ρ_s to each accepted student. The first condition states that the number of students whose rationed demand (henceforth, simply “demand”) is a subset of S' exceeds the total maximum for S' . The second condition adds that, for each subset S'' of S' , the students from condition 1 whose demand intersects S'' also exceed the maximum for S'' . Among all sets in excess demand at (ρ_z, Q_z) , let $E^*(\rho_z, Q_z)$ be one with maximal cardinality. Observe that $E^*(\rho_z, Q_z)$ turns out to be unique and can be found in a computationally efficient way (Andersson, Andersson, and Talman 2013). Again, for simplicity, we suppress arguments and write only E_z^* .

Algorithm:

Time begins at $z = 1$. The distribution is initialized at $\rho_1 = (1, \dots, 1)$ and Q_1 such that for each $i \in N$, we have $Q_{i1} = S$. The algorithm consists of three subroutines, described below. In each case, assume for notational simplicity that the subroutine is entered at time z . The entry time for any subroutine is always the exit time of the previous; time only passes during subroutines. The algorithm begins in the subroutine PAUSE.

Pause: Calculate E_z^* .

If E_z^* is empty, the algorithm is terminated.

If all schools in E_z^* can be decremented, i.e., if each $s \in E_z^*$ has $\rho_{sz} > b_s$, then go to subroutine DECREMENTING.

If E_z^* contains a school that cannot be decremented, go to subroutine rejection.

Regardless of the condition under which the algorithm exits this subroutine, the time at exit equals the time at entry; the PAUSE subroutine is instantaneous.

Rejection: Let $s \in E_z^*$ have $\rho_{sz} = b_s$. If there are several such s , choose one arbitrarily. Find the $i \in N$ who is $\succ_s$ -lowest priority among those students j with $s \in D_{jz}$. Advance time one unit to $z + 1$, and let $\rho_{z+1} = \rho_z$. For each $j \in N$ with $i \succ_s j$ or $j = i$, let $Q_{j,z+1} = Q_{jz} \setminus \{s\}$; for each other j , let $Q_{j,z+1} = Q_{jz}$. Go to subroutine PAUSE.

Decrementing: Advance time continuously leaving Q_z unchanged, and decrement ρ_z so that for $z' > z$,

$$\rho_{sz'} = \begin{cases} \frac{\rho_{sz} z}{z'} & s \in E_z^*, \text{ and} \\ \rho_{sz} & s \notin E_z^*. \end{cases}$$

Exit at time $z' > z$, when either

- (1) there is $s \in E_z^*$ with $\rho_{sz'} = 1/k$ for $k \in \mathbb{N}$,²¹ or
- (2) there is i such that $D_{iz} \subseteq E_z^*$ at z for whom $D_{iz'} \neq D_{iz}$.

Upon exit, go to subroutine PAUSE.

We make several remarks on the algorithm. First, note that s will reject at most $|N| - b_s^{-1}$ times before it is no longer in excess demand, so subroutine REJECTION will be invoked at most $|S|(|N| - b_s^{-1})$ times. Second, consider the DECREMENTING subroutine. The linear preference domain yields that for each $z' > z > 0$,

$$(1) \quad \rho_{sz} v_i(s) \geq \rho_{tz} v_i(t) \Leftrightarrow \rho_{sz'} v_i(s) = \frac{z \rho_{sz}}{z'} v_i(s) \geq \frac{z \rho_{tz}}{z'} v_i(t) = \rho_{tz'} v_i(t).$$

Thus, if i 's demand falls within the set of excess demand, i.e., if $D_{iz} \subseteq E_z^*$, then their demand is constant for a positive amount of time in the algorithm, making the subroutine well defined. Moreover, assuming exit condition 1 is not reached, exit condition 2 says that DECREMENTING exits precisely when one of these students experiences an enlargement of demand. Because of these features, we say that demand is not elicited during decrementing but that, similar to a clock auction, some participant will stop the clock at z' to notify the auctioneer that a change has occurred and demand needs to be reelicited. Finally, when the algorithm terminates, say at time $\bar{z}$, there will be no sets in excess demand, and so Hall's theorem (Hall 1935) guarantees the existence of an equilibrium matching σ , which, of course, corresponds to the existence of allocation $(\rho_{\bar{z}}, \sigma)$.

In online Appendix C we provide a calculation of the algorithm for a small example problem, and we finish the proof of Theorem 5.

THEOREM 5: *For linear preference profiles from the NCBI domain, the algorithm terminates in finite time with finitely many elicitations of demand and results in a maximal RCE.*

VII. Discussion

A. Dropping Inferior Empty Schools

Let an allocation be a weak RCE if it satisfies *fairness*, *exhaustiveness*, and simply that each empty $s \in S$ have $\rho_s = 1$. Thus, we replace *inferior empty schools* with a more standard Walrasian-type condition. Clearly, as RCEs are weak RCEs, these exist on the NCBI domain, but Proposition 2 does not hold, as trading paths between them can terminate by adding a student to an empty school. It follows that Theorem 2 also fails. However, we can show that this only happens for paths on which every student is indifferent between the two equilibria. Thus, the result for

²¹ This implies that at time z' , the capacity of school s increases by one to k , compared to its capacity at time z .

welfare-improving and welfare-harming trading cycles holds, which is all that is required to show the upper lattice. Of course, *strategy-proofness* and the algorithm can be directly carried over.

B. Applications beyond School Choice

The first application is to the problem of *labor markets with financially constrained start-ups*. Consider the labor market where there is a finite set of start-ups F and a finite set of workers W . Each start-up $f \in F$ is subject to some (hard) financial constraints so that expenditure for labor employment $\kappa_f > 0$ is fixed for the modeled time period. Given this, the start-up selects employees following a priority order $\succ_f$. The labor market is protected by a minimum wage $\underline{w}$ such that $0 < \underline{w} \leq \kappa_f$ for each $f \in F$. Each worker $w \in W$ has a complete, transitive, and continuous preference relation R_w over $\{(t, f) \in \mathbb{R} \times F : 0 \leq t \leq \kappa_f\}$ such that for each $f \in F$ and each $t, t' \in [0, \kappa_f]$ with $t > t'$, $(t, f) P_w(t', f)$. The tuple $(F, W, \mathbf{R}, \kappa, \succ, \underline{w})$ summarizes primitives of the problem.

We reformulate the problem to apply our results. Let F , W , and $\succ$ be as defined above. For each $f \in F$, let b_f be such that $b_f^{-1} = \kappa_f / \underline{w} \in \mathbb{N}$. Each worker $w \in W$ has a complete, transitive, and continuous preference relation R_w^* over $[0, 1] \times F$ such that for each $f, f' \in F$ and each $t, t' \in [0, 1]$, $(t, f) R_w^*(t', f')$ if and only if $(t \kappa_f, f) R_w(t' \kappa_{f'}, f')$. It is easy to see that all our results hold in the problem $(F, W, \mathbf{R}^*, \mathbf{b}, \succ)$, and therefore, the insights can be carried over to the original problem $(F, W, \mathbf{R}, \kappa, \succ, \underline{w})$.

The second application is to the problem of *allocating polluting firms*. Instead of students and schools, consider polluting firms and subnations. Each firm is willing to invest in at most one subnation. Each subnation s is endowed with an amount of the same environmental resource, and the priority for selecting polluting firms is based on their industrial development policy. Let b_s^{-1} be the maximum number of polluting firms that the subnation is willing to admit. Each firm prefers to locate to a subnation where they are allowed a higher level of pollution. As this is simply a reinterpretation of the model, we are able to directly apply our results for this application.

VIII. Conclusion

We provide a new framework to model school choice with crowding and establish analogs of key results in the school choice and matching literature. Many issues of perennial interest—diversity considerations, efficiency improvements, sibling guarantees, etc.—can be reconsidered in this model. We also align with the empirical literature's explicit consideration of per capita crowding metrics.

The model also opens up completely new avenues of exploration not suited to the standard one. Most interesting is the question of redistributing resources *between* schools. We have assumed that each school has a fixed endowment of resources, but in practice, the resources of all schools in a given district are usually controlled centrally by that district. For example, money or physical infrastructure (such as a trailer) can be shifted from one school to another; counselors routinely serve multiple schools. Thus, we can design new mechanisms encompassing such distribution

between schools. We can also ask if there are ways to redistribute resources so that, ex post, our mismatch between supply and demand is completely eliminated. More generally, we can study comparative statics in this dimension and build a more dynamic model of educational resource allocation.

It is worth noting, in conclusion, that the model implicitly covers a wide variety of different crowding effects. We have modeled crowding in a hyperbolic way: n students share resources at (approximately) $1/n$ per capita. However, our relatively unrestricted domain of preferences contains within it many subdomains that each embody different notions of crowding. For example, the reader may verify that a function of the form $(\rho_s, s) \mapsto v(s) \frac{\rho_s^{1-\eta}}{1-\eta}$ exhibits increasing marginal disutility for crowding if $\eta > 2$.²²

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²² Write $\rho_s = 1/n$, and calculate $\frac{d^2}{dn^2} u(1/n, s)$.

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